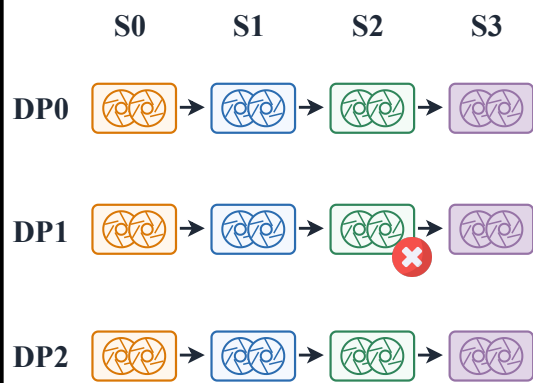


1 Training Job

Sparse MoE Distributed
Training (3 DP $\times$ 4 Pipeline)

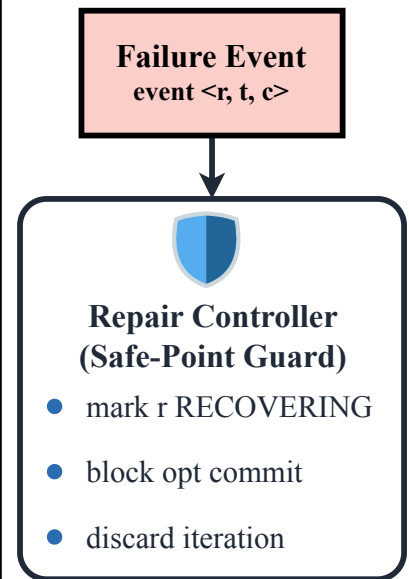


 Healthy worker

 Failed worker (rank r)

→ Control Flow

② Failure Detection + Safe-Point Repair (R1)



- r** : failed rank
- t** : failure step (current)
- c** : checkpoint step

→ Path P (Peer)

3 Guarded Policy (R2)

Policy Metric

$$\Phi'(t) = \frac{S(t) + |E_{\text{new}}| \cdot \Delta}{N_{\text{expert}} \cdot W}$$

Inputs

- $\Delta = t - c$
- peerAvail
- $S(t)$
- E_{new}

Thresholds

- $\Delta_{\min}$
- $\Delta_{\max}$
- $\Phi_{\max}$

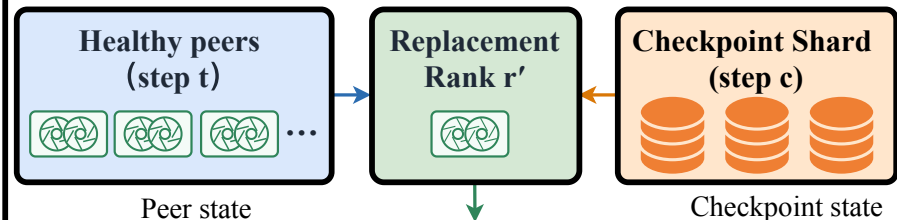
YES

guards pass?
HYBRID ?

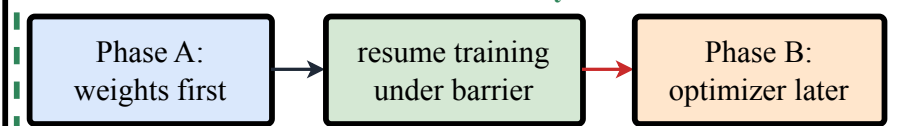
NO

→ Path C (Checkpoint)

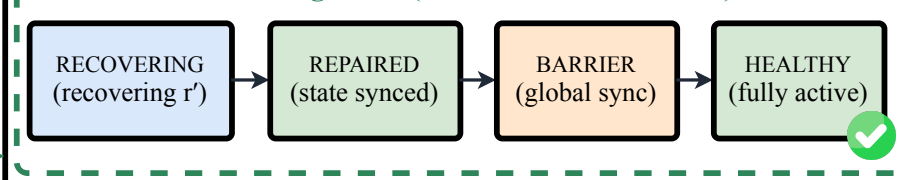
4a Hybrid Recovery (fast path)



Two-Phase Recovery Protocol



Reintegration (Guarded State Machine)



4b Checkpoint Restart (fallback)



Hybrid Path

→ Restart Path

5 Observability Logs (R3)



Auditable recovery decisions



policy inputs +
reason

TTTR / TTTFR
latencystate-machine
trace

quality metrics
(loss, ppl,
balance)

R1/R2/R3: MoEGambit Rules